Part 1: Kubernetes Troubleshooting

Deployment manifest:

Issue 1: There is a mismatch for the application labels in spec.selector.matchLabels.app and labels.application. It should be either web-app or webapp everywhere otherwise the deployment won't match its pods.

Issue 2: Annotation values for Prometheus should be strings so prometheus.io/scrape value t rue should be "true".

Issue 3: Wrong port and protocol for nginx. The protocol should be TCP and the port 80.

Issue 4: Incorrect resource range – requests.memory is 64Mi meanwhile limits.memory is 32Mi. Limit must be greater or equal than request.

Issue 5: DATABASE_PASSWORD should come from a Secret manifest kind and not be hardcoded.

Issue 6: VolumeMount references data but only cache is defined in volumes.

Service manifest:

Issue 1: There is a namespace mismatch, service is in staging but everything else is in production so the ingress can't route to it.

Issue 2: The selector for spec.selector.app does not match the deployment pod labels.

Ingress manifest:

Issue 1: Missing required pathType which in this case should be Prefix because of the path.

Issue 2: It points to a non-existent service name webapp-svc. It should be web-app-service.

NetworkPolicy manifest:

Issue 1: Ingress here allows only traffic on port 443 but the app runs on HTTP (80). This blocks traffic.

PDB manifest:

Issue 1: You cannot set both minAvailable and maxUnavailable because if you set minAvailable to 3 that means 0 pods may go down and if you set maxUnavailable to 1 that means 1 pod may go down and both things can't happen at the same time meaning it is a contradiction.

HPA manifest:

Issue 1: scaleTargetRef.name: webapp but Deployment is web-app.

Issue 2: minReplicas == maxReplicas == 3 meaning HPA won't scale at all.